Open Cisco Packet Tracer and set up a simple network with two routers connected via a serial cable, assigning IP addresses to both router interfaces.

ANS:

To set up two routers in Cisco Packet Tracer with a serial cable and configure their IP addresses:

1. **Add Hardware:** Place two routers (`Router0` and `Router1`) and install **HWIC-2T** serial interface modules on both after powering them off.
2. **Connect Cable:** Use a **Serial DCE** cable to connect `Serial0/0/0` on Router0 to `Serial0/0/0` on Router1.
3. **Configure Router0 (CLI):**
4. Plaintext

```
None
enable
configure terminal
interface serial 0/0/0
ip address 192.168.1.1 255.255.255.0
clock rate 64000
no shutdown
```

5.
6. **Configure Router1 (CLI):**
7. Plaintext

```
None
enable
configure terminal
interface serial 0/0/0
ip address 192.168.1.2 255.255.255.0

   8. no shutdown
```

9.

To configure a static route on **Router1** to reach Router2's LAN network (e.g., `192.168.2.0/24`) and test connectivity:

Step 1: Add the Static Route on Router1

Click on **Router1**, open the **CLI** tab, and enter the following commands (replace `192.168.1.2` with Router2's connected interface IP if different):

Plaintext

```
None
enable
configure terminal
ip route 192.168.2.0 255.255.255.0 192.168.1.2
exit
```

- **Verification:** Run `show ip route` and ensure a static route entry (`S`) for `192.168.2.0/24` is visible in the routing table.

Step 2: Test Connectivity with Ping

From Router1's CLI (in privileged EXEC mode), ping the PC's IP address on Router2's LAN:

Plaintext

```
None
ping 192.168.2.10
```

- **Verification:** The command returns success indicator exclamation points (`!!!!!`) showing a successful response rate.

To connect your Ahmedabad branch (Router1) and Surat branch (Router2) so their LAN networks can communicate, configure static routes pointing to each other's subnets across the serial link.

Assuming the following network scheme:

- **Serial Link (Between Routers):** `192.168.1.0/30` (Router1: `192.168.1.1`, Router2: `192.168.1.2`)
- **Ahmedabad LAN (Router1):** `192.168.10.0/24`
- **Surat LAN (Router2):** `192.168.20.0/24`

Step 1: Configure Ahmedabad Router (Router1)

Open **Router1**, go to the **CLI** tab, and add a static route pointing to the Surat LAN via Router2's serial IP:

Plaintext

```
None
enable
configure terminal
ip route 192.168.20.0 255.255.255.0 192.168.1.2
exit
```

- **Verification:** Run `show ip route` and confirm the static route (`S`) for `192.168.20.0/24` appears.

Step 2: Configure Surat Router (Router2)

Open **Router2**, go to the **CLI** tab, and add a static route pointing to the Ahmedabad LAN via Router1's serial IP:

Plaintext

```
None
enable
configure terminal
ip route 192.168.10.0 255.255.255.0 192.168.1.1
exit
```

- **Verification:** Run `show ip route` and confirm the static route (`S`) for `192.168.10.0/24` appears.

Step 3: Test Cross-Branch Connectivity

From a PC on the Ahmedabad LAN, open the command prompt and ping a PC on the Surat LAN:

Plaintext

```
None
ping 192.168.20.10
```

- **Verification:** Successful replies (`Reply from 192.168.20.10...`) indicate full routing between both branch offices.

List two advantages and two disadvantages of static routing compared to dynamic routing, using examples from real-life apps like IRCTC or Swiggy to illustrate your points.

Hint: Think about how static routes might impact delivery or booking updates in these apps.
Ans:

Here are two key advantages and two disadvantages of static routing compared to dynamic routing, illustrated through everyday scenarios in apps like Swiggy and IRCTC.

Advantages of Static Routing

1. **Higher Security and Complete Control**
    - **Why it matters:** Static routes do not share routing information with external networks, meaning hackers cannot spoof routing updates or inject malicious paths into the routing table.
    - **Swiggy/IRCTC Example:** IRCTC uses static, hardcoded connections for critical payment gateway transactions between specific bank servers and central booking databases to ensure maximum security against route manipulation.
2. **Zero Resource Overhead (CPU and Bandwidth)**
    - **Why it matters:** Routers do not waste processing power running complex algorithms (like Dijkstra's in OSPF) or bandwidth sending periodic "hello" update packets to neighboring routers.

- **Swiggy/IRCTC Example:** For smaller regional hubs or edge dark stores with limited processing capacity, static routing keeps CPU and memory entirely dedicated to handling high-speed food delivery tracking.

Disadvantages of Static Routing

1. **No Automatic Failover (Zero Redundancy)**
   - **Why it matters:** If a primary link goes down, static routes cannot automatically reroute traffic through an alternative path. Human intervention is required to update the routing table.
   - **Swiggy/IRCTC Example:** If the direct fiber link between your Ahmedabad and Surat Swiggy branches experiences a physical cut, static routing means branch communication freezes entirely until a network engineer manually redirects the traffic. Dynamic routing, by contrast, would instantly reroute packets via an alternate path (like Vadodara).
2. **Poor Scalability and High Administrative Burden**
   - **Why it matters:** Every time a new network segment, branch, or dark store is added, an administrator must manually log into every single router to update the routes.
   - **Swiggy/IRCTC Example:** When Swiggy expands into 50 new tier-2 cities overnight, configuring static routes for every single delivery hub across the entire nationwide network becomes an unmanageable, error-prone administrative nightmare.

List three differences between static routing and dynamic routing in a table format, using real-world analogies from apps like Google Maps or Swiggy to make each point relatable.

ANS:

Here are three key differences between static routing and dynamic routing, compared using relatable analogies from Google Maps and Swiggy:

Feature	Static Routing	Dynamic Routing	Real-World Analogy (Swiggy / Google Maps)
Adaptability & Failover	**Rigid:** Routes are manually hardcoded; if a link fails, traffic drops until a human manually updates it.	**Flexible:** Automatically detects network changes and calculates backup paths in real-time.	**Google Maps** instantly rerouting a Swiggy delivery partner around a road closure vs. following a rigid, outdated paper map that leads straight into traffic.
Scalability & Setup	**Manual & Labor-Intensive:** Requires an administrator to configure routes on every single router whenever a network grows.	**Automated:** Routers dynamically discover and share network information with zero manual intervention.	**Swiggy** automatically onboarding hundreds of new restaurant partners nationwide vs. manually dispatching updated paper address books to every hub.

Resource Overhead	**Zero Overhead:** Consumes no CPU power or background bandwidth for routing calculations.	**High Overhead:** Constantly consumes CPU and bandwidth to send status updates and run path algorithms.	A delivery executive memorizing a fixed shortcut to a nearby apartment vs. running a heavy live-tracking navigation app that constantly consumes battery and data.

Open Cisco Packet Tracer and load a pre-configured network file with three routers using RIP. Use the 'show ip route' and 'show ip protocols' commands on each router to observe and write down the RIP-learned routes and their metrics.

ANS:

Step 1: Access the Router CLI

Click on each router in your workspace, navigate to the **CLI** tab, press **Enter**, and enter privileged EXEC mode:

Plaintext

```
None
enable
```

Step 2: Check RIP Routes (`show ip route`)

Run the routing table command to inspect how routes are learned:

Plaintext

```
None
show ip route
```

- **What to look for:** Look for entries preceded by the letter `R`, which stands for RIP.
- **Reading the Metric:** Next to the route, you will see a bracketed value like `[120/1]`.
  - `120` is the **Administrative Distance (AD)** for RIP.
  - `1` is the **Metric** (hop count), representing the number of routers the packet must pass through to reach the destination network.

Step 3: Check RIP Configuration & Timers (`show ip protocols`)

Run the protocol status command to verify RIP settings, advertised networks, and timers:

Plaintext

```
None
show ip protocols
```

- **What to look for:** Verify that **Routing Information Protocol (RIP)** is listed as active.

Simulate a network change by shutting down one interface on a router in the RIP-enabled topology. Observe and record how long it takes for the routing tables on the other routers to update and remove the unreachable route.

Hint: Use the 'shutdown' command on an interface and monitor with 'show ip route' repeatedly.

ANS:

To simulate a network failure and observe RIP convergence time, follow these steps to shut down an interface and track the routing table updates:

1.Shut Down the Interface: Step 1.

Log into **Router1** (or whichever router you are testing), enter global configuration mode, and shut down the active serial interface:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

shutdown

exit
```

Verification: Running show ip interface brief confirms the interface status is administratively down.

2.Start a Stopwatch and Monitor Routing Tables: Step 2.

Immediately switch to a neighboring router (e.g., Router2 or Router3), start a stopwatch, and repeatedly execute the routing table command every 10 to 15 seconds:

Plaintext

```
None
show ip route
```

Verification: Watch for the specific RIP-learned route (marked with an R) to either vanish or trigger an update.

3.Record the Final Convergence Time: Step 3.

Stop the timer the exact moment the unreachable route is completely removed from the routing table.

Verification: Under standard RIP default settings, it takes **180 seconds** (the invalid timer) for a route to be marked as invalid/dead, and **240 seconds** (the flush timer) for it to be completely purged from the routing table.

Explain in 3-4 sentences how dynamic routing in apps like Uber or Zomato delivery tracking is similar to RIP dynamic routing in networks. Focus on how routes or paths are updated automatically in both cases.

ANS:

- **Continuous Monitoring:** RIP routers exchange periodic status messages, just as delivery apps track live GPS pings from drivers.
- **Automatic Rerouting:** When a network link goes down or a road gets blocked, both systems instantly recalculate and switch to alternative paths without manual intervention.
- **Resilience:** Both ensure data packets or food deliveries successfully navigate around obstacles to reach their destination.

Set up a network topology with three routers in Cisco Packet Tracer, connect them using serial or Ethernet links, and assign IP addresses to all interfaces.

ANS:

Step 1: Place and Equip the Routers

1. Place three modular routers (`Router0`, `Router1`, `Router2`) on the workspace.
2. Click on each router, go to the **Physical** tab, turn off the power switch, insert the **HWIC-2T** serial module into an empty slot, and turn the power back on.

- **Verification:** Serial ports (e.g., `Serial0/0/0`) appear in the physical view for all three devices.

Step 2: Connect via Serial Cables

Using a **Serial DCE** cable:

- Connect **Router0** (`Serial0/0/0`) to **Router1** (`Serial0/0/0`).
- Connect **Router1** (`Serial0/0/1`) to **Router2** (`Serial0/0/0`).
- **Verification:** Link lights turn amber and then green once configured.

Step 3: Configure Router0 (CLI)

Open **Router0** and assign its serial interface IP (`192.168.12.1/30`):

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.12.1 255.255.255.252

clock rate 64000

no shutdown

exit
```

Step 4: Configure Router1 (CLI)

Open **Router1** and configure both connected serial interfaces:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0
```

```
ip address 192.168.12.2 255.255.255.252

no shutdown

interface serial 0/0/1

ip address 192.168.23.1 255.255.255.252

clock rate 64000

no shutdown

exit
```

Step 5: Configure Router2 (CLI)

Open **Router2** and assign its serial interface IP (`192.168.23.2/30`):

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.23.2 255.255.255.252

no shutdown

exit
```

- **Verification:** Run `show ip interface brief` on each router to confirm all active serial interfaces show status `up` and protocol `up`.

Enable RIP version 2 on all three routers in your topology, advertise all directly connected networks, and verify that each router learns routes to all subnets using the 'show ip route' command.

ANS:

To enable RIP version 2 and advertise your connected networks across all three routers, follow these configuration steps:

Step 1: Configure Router0

Open **Router0**, go to the CLI, and enter the RIPv2 configuration commands:

Plaintext

```
None
enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.12.0

exit
```

- **Verification:** Running `show ip protocols` confirms RIP version 2 is running and network `192.168.12.0` is advertised.

Step 2: Configure Router1

Open **Router1**, go to the CLI, and configure RIPv2 for both connected subnets:

Plaintext

```
None
enable

configure terminal
```

```
router rip

version 2

no auto-summary

network 192.168.12.0

network 192.168.23.0

exit
```

- **Verification:** Router1 will act as the intermediary hub, exchanging updates with both Router0 and Router2.

Step 3: Configure Router2

Open **Router2**, go to the CLI, and configure RIPv2 for its connected subnet:

Plaintext

```
None
enable

configure terminal

router rip

version 2

no auto-summary

network 192.168.23.0

exit
```

- **Verification:** Running `show ip protocols` on Router2 confirms it is actively listening for RIPv2 updates on `192.168.23.0`.

Step 4: Verify Routing Tables

Go back to **Router0** (and subsequently Router1 and Router2) and inspect the routing table:

Plaintext

```
None
show ip route
```

- **Verification:** Look for entries marked with an **R**. Router0 should now successfully show a route to the remote `192.168.23.0/30` network learned via RIP with a hop count metric of `[120/2]`.

Change the hop count metric for one of the networks by adding an extra router between two existing routers, then observe and record how the hop count value changes in the routing tables.

ANS:

To observe how adding an intermediate router alters the RIP hop count metric, modify your topology by inserting a new router (Router3) between **Router0** and **Router1**.

Step 1: Insert Router3 and Re-cable

1. Drag a new modular router (**Router3**) onto the workspace and insert an **HWIC-2T** module (remember to power down first).
2. Delete the direct serial link between **Router0** and **Router1**.
3. Connect **Router0** (Serial0/0/0) to **Router3** (Serial0/0/0) using a Serial DCE cable.
4. Connect **Router3** (Serial0/0/1) to **Router1** (Serial0/0/0) using a Serial DCE cable.

- **Verification:** The physical topology now shows the path Router0 <--> Router3 <--> Router1 <--> Router2.

Step 2: Configure Router3 (Interfaces & RIPv2)

Assign IP addresses to Router3's interfaces (192.168.13.0/30 for Router0 link, 192.168.31.0/30 for Router1 link) and enable RIPv2:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.13.2 255.255.255.252

no shutdown

interface serial 0/0/1

ip address 192.168.31.1 255.255.255.252

clock rate 64000

no shutdown

exit


router rip

version 2

no auto-summary

network 192.168.13.0

network 192.168.31.0

exit
```

- **Verification:** Running show ip interface brief shows both serial interfaces on Router3 as up/up.

Step 3: Update Router0 and Router1 IP Settings

Adjust the connected interface IPs on **Router0** and **Router1** to match the new subnets, then update their RIP network advertisements:

On Router0:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.13.1 255.255.255.252

clock rate 64000

no shutdown

exit

router rip

version 2

no auto-summary

network 192.168.13.0

no network 192.168.12.0

exit
```

On Router1:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0
```

```
ip address 192.168.31.2 255.255.255.252

no shutdown

exit

router rip

version 2

no auto-summary

network 192.168.31.0

no network 192.168.12.0

exit
```

Step 4: Observe the Updated Hop Count Metric

Wait 30 seconds for RIP convergence, then check the routing table on **Router0**:

Plaintext

```
None
show ip route
```

Expected Metric Change:

- **Before modification:** Route to 192.168.23.0/30 on Router0 was learned directly via Router1 with metric **[120/2]** (2 hops: Router1 → Router2).
- **After modification:** Route to 192.168.23.0/30 on Router0 now shows metric **[120/3]** because the path traverses 3 router hops (Router3 → Router1 → Router2).
- **Verification:** The entry for 192.168.23.0/30 explicitly displays [120/3] in the show ip route output on Router0.

Use the simulation mode in Cisco Packet Tracer to capture and analyze RIP update packets as they are exchanged between routers.

Hint: Filter for RIP packets in the simulation panel and note the source, destination, and advertised routes in each update.

ANS:

Here is how to use Simulation Mode in Cisco Packet Tracer to capture, filter, and analyze live RIP update packets exchanged between your routers.

1.Switch to Simulation Mode: Step 1.

Click the **Simulation Mode** icon (the stopwatch/clock symbol) located in the bottom-right corner of the workspace, next to the Realtime tab.

Verification: The workspace background turns slightly orange, and the simulation control panel appears.

2.Filter for RIP Packets: Step 2.

1. In the Simulation Panel, click **Edit Filters** (or **Show All/None**).
2. Check **IPv4**, then click the arrow next to it or open the **UDP** / miscellaneous filters to select **RIP**. (Alternatively, click *Edit Filters*, go to the *Misc* tab, and select **RIP**).
   Verification: Only RIP traffic (typically using UDP port 520) will be captured and displayed in the event list.

3.Trigger or Wait for a RIP Update: Step 3.

RIP routers send updates every 30 seconds. To force an immediate update without waiting:

1. Go to any router's CLI and run:

Plaintext

```
None
clear ip route *
```

2. Click the **Capture / Forward** (play button) or **Auto Capture / Play** button in the Simulation Panel.
   Verification: Colored envelope icons (representing RIP update packets) appear moving across the serial links between the routers.

4.Inspect the RIP Packet Details: Step 4.

1. Click on one of the captured envelope icons on the canvas (or click the colored box under the **Info** column in the event list).

2. In the **PDU Information** window that pops up, click the **Layer 3** and **Layer 4** tabs to view source/destination IPs and UDP ports.
3. Click the **RIP** tab to inspect the payload, including the **Command** (1 for Request, 2 for Response), **Version** (Version 2), **IP Address** prefixes, and associated **Metric** (hop counts).
   Verification: The RIP table inside the packet details lists the exact subnets and metric values currently advertised by that router.

Open Cisco Packet Tracer and create a simple network with three routers connected in a triangle topology using serial or Ethernet links.

ANS:

Here is how to set up three routers in a triangle topology in Cisco Packet Tracer and assign IP addresses to their connected interfaces.

1.Place and Arrange the Routers: Step 1.

Place three modular routers (such as the 2911 model) on the workspace canvas arranged in a triangle shape: **Router0** at the top, **Router1** at the bottom-left, and **Router2** at the bottom-right.

Verification: All three router icons are visible on the workspace.

2.Install Serial Interface Modules: Step 2.

Most default routers lack serial ports. Click on each router, go to the **Physical** tab, turn off the power switch, drag an **HWIC-2T** serial module into an empty slot, and turn the power back on. Repeat for Router1 and Router2.

Verification: Serial ports (Serial0/0/0, Serial0/0/1) appear in the physical view for all three devices.

3.Connect the Routers in a Triangle: Step 3.

Select the **Serial DCE** cable from the connections menu and link the routers to form a closed loop:

- Connect **Router0** (Serial0/0/0) to **Router1** (Serial0/0/0)
- Connect **Router1** (Serial0/0/1) to **Router2** (Serial0/0/0)
- Connect **Router2** (Serial0/0/1) to **Router0** (Serial0/0/1)
  Verification: Link lights between the routers turn amber and then green once configured.

4.Configure Router0 Interfaces: Step 4.

Open **Router0**, go to the CLI, and assign IP addresses for its two serial links (using /30 subnets):

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.12.1 255.255.255.252

clock rate 64000

no shutdown

interface serial 0/0/1

ip address 192.168.30.2 255.255.255.252

clock rate 64000

no shutdown

exit
```

Verification: Running show ip interface brief shows both serial interfaces as up/up.

5.Configure Router1 and Router2 Interfaces: Step 5.

Open **Router1** and **Router2** to configure their respective connecting links:

On Router1:

Plaintext

```
None
enable

configure terminal
```

```
interface serial 0/0/0

ip address 192.168.12.2 255.255.255.252

no shutdown

interface serial 0/0/1

ip address 192.168.23.1 255.255.255.252

clock rate 64000

no shutdown

exit
```

On Router2:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.23.2 255.255.255.252

no shutdown

interface serial 0/0/1

ip address 192.168.30.1 255.255.255.252

no shutdown

exit
```

Assign IP addresses to all router interfaces in your Packet Tracer network and verify connectivity using the ping command.

ANS:

Here is how to verify that IP addresses are correctly assigned across all interfaces in your triangle network and test connectivity using the `ping` command.

Step 1: Verify Interface Status on All Routers

Open each router's CLI in privileged EXEC mode to confirm that all serial interfaces are active and assigned to their correct `/30` subnets:

Plaintext

```
None
show ip interface brief
```

- **Verification:** All connected serial interfaces (`Serial0/0/0` and `Serial0/0/1`) must display a status of **up** and protocol **up**.

Step 2: Test Direct Link Connectivity with Ping

Because routers can directly communicate with their immediate neighbors on each serial link, run the following `ping` commands from each router's CLI:

On Router0:

Ping Router1 across the first link and Router2 across the second link:

Plaintext

```
None
ping 192.168.12.2

ping 192.168.30.1
```

- **Verification:** Successful pings return success indicator exclamation points (`!!!!!`) with a 100% success rate.

On Router1:

Ping Router0 and Router2:

Plaintext

```
None
ping 192.168.12.1

ping 192.168.23.2
```

On Router2:

Ping Router1 and Router0:

Plaintext

```
None
ping 192.168.23.1

ping 192.168.30.2
```

- **Verification:** Receiving successful replies from both neighboring interfaces on every router confirms that all IP addresses are properly configured and physical links are operational.

Configure OSPF on each router using the router ospf 1 command and the network command to advertise all directly connected networks.

ANS:

To configure OSPF on all three routers in your triangle topology and advertise their directly connected networks using Area 0, follow these CLI configuration steps:

1.Configure Router0: Area 0 OSPF Setup.

Open **Router0**, enter global configuration mode, start OSPF process 1, and advertise its two connected serial links (192.168.12.0/30 and 192.168.30.0/30) using wildcard masks:

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.12.0 0.0.0.3 area 0

network 192.168.30.0 0.0.0.3 area 0

exit
```

Verification: Running show ip protocols confirms OSPF process 1 is active and routing updates are enabled for both networks.

2.Configure Router1: Area 0 OSPF Setup.

Open **Router1**, enter global configuration mode, start OSPF process 1, and advertise its connected links (192.168.12.0/30 and 192.168.23.0/30):

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.12.0 0.0.0.3 area 0

network 192.168.23.0 0.0.0.3 area 0

exit
```

Verification: The console outputs an OSPF neighbor adjacency notification (e.g., %OSPF-5-ADJCHG: Process 1, Nbr 192.168.12.1 on Serial0/0/0 from LOADING to FULL, Loading Done).

3.Configure Router2: Area 0 OSPF Setup.

Open **Router2**, enter global configuration mode, start OSPF process 1, and advertise its connected links (192.168.23.0/30 and 192.168.30.0/30):

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.23.0 0.0.0.3 area 0

network 192.168.30.0 0.0.0.3 area 0

exit
```

Verification: Run show ip route on any router to verify that OSPF-learned routes (preceded by **O**) appear for all subnets across the triangle loop.

Observe and document the OSPF adjacency formation process by checking the output of the show ip ospf neighbor command on each router.

Hint: Look for the FULL/DR or FULL/BDR state to confirm adjacency.

ANS:

To observe the OSPF adjacency formation process and confirm full synchronization across your triangle topology, execute the neighbor status command on each router CLI.

Step 1: Check Router0

Open **Router0**, go to the CLI, and run:

Plaintext

```
None
show ip ospf neighbor
```

- **Verification:** Look for the entries corresponding to Router1 and Router2. Confirm the state reads **`FULL`** (e.g., `FULL/DROther` or `FULL/-` on serial point-to-point links), indicating that databases are fully synchronized.

Step 2: Check Router1

Open **Router1**, go to the CLI, and run:

Plaintext

```
None
show ip ospf neighbor
```

- **Verification:** Confirm that neighbor Router0 and Router2 both display a **`FULL`** adjacency state along with their respective Router IDs, interface names, and dead timers.

Step 3: Check Router2

Open **Router2**, go to the CLI, and run:

Plaintext

```
None
show ip ospf neighbor
```

- **Verification:** Verify that all connected adjacencies show **`FULL`**, confirming that the triangle loop is fully converged and operational.

Change the OSPF cost on one interface using the ip ospf cost command and verify how it affects the OSPF routing table.

Hint: Use show ip route ospf to see route changes after modifying the cost.

ANS:

To change the OSPF cost on a specific interface and observe how it alters the preferred path in your network, follow these steps:

1.Change the OSPF Cost on Router0: Interface Configuration.

Open **Router0**, enter global configuration mode, select the serial interface connecting to Router1, and assign a custom OSPF cost:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip ospf cost 50

exit
```

Verification: Run show ip ospf interface serial 0/0/0 and verify that the interface cost explicitly displays as 50.

2.Inspect OSPF Routes: Routing Table Check.

Switch to **Router2**, wait a few seconds for OSPF to recalculate the Dijkstra algorithm, and view only the OSPF-learned routes:

Plaintext

```
None
show ip route ospf
```

Verification: Notice how the cumulative metric (cost) for subnets accessed via that path increases to reflect the newly added cost of 50.

3.Observe Alternative Path Selection: Path Recalculation Analysis.

Because OSPF always chooses the path with the lowest total cost, if the modified path's cost now exceeds the alternative route across the triangle loop, OSPF will dynamically switch the routing table to use the cheaper link.

Verification: Check the next-hop gateway IP address in the routing table entry to confirm if traffic has automatically shifted to the alternate path.

Using Cisco Packet Tracer, create a network topology with three routers in two OSPF areas (Area 0 and Area 1), and assign appropriate IP addresses to all interfaces.

ANS:

Here is how to set up a 3-router topology split across two OSPF areas (**Area 0** as the backbone and **Area 1** as a non-backbone area) in Cisco Packet Tracer, along with interface IP assignments.

Network Addressing Plan

- **Area 0 (Backbone):**
  - Router0 to Router1 Link: `192.168.12.0/30` (Router0: `192.168.12.1`, Router1: `192.168.12.2`)
  - Router0 LAN: `192.168.10.0/24` (Router0: `192.168.10.1`)
- **Area 1 (Non-Backbone):**
  - Router1 to Router2 Link: `192.168.23.0/30` (Router1: `192.168.23.1`, Router2: `192.168.23.2`)
  - Router2 LAN: `192.168.20.0/24` (Router2: `192.168.20.1`) *(Note: **Router1** acts as the Area Border Router [ABR], connecting Area 0 and Area 1).*

Step 1: Place and Equip Routers

1. Place three modular routers (`Router0`, `Router1`, `Router2`) on the workspace.
2. Click each router, go to the **Physical** tab, power off the device, insert an **HWIC-2T** serial module into an empty slot, and power it back on.
- **Verification:** Serial ports (`Serial0/0/0`, `Serial0/0/1`) appear in the physical view.

Step 2: Connect the Topology

Using a **Serial DCE** cable:

- Connect **Router0** (`Serial0/0/0`) to **Router1** (`Serial0/0/0`).

- Connect **Router1** (`Serial0/0/1`) to **Router2** (`Serial0/0/0`).
- *(Optional)* Connect a switch and PC to Router0's and Router2's LAN interfaces for end-to-end testing.

Step 3: Configure Router0 (Area 0)

Open **Router0**, go to the CLI, and assign IP addresses to its serial and LAN interfaces:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.12.1 255.255.255.252

clock rate 64000

no shutdown

interface gigabitethernet 0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

exit
```

Step 4: Configure Router1 (ABR - Area 0 & Area 1)

Open **Router1**, go to the CLI, and configure both connected serial interfaces:

Plaintext

```
None
enable

configure terminal
```

```
interface serial 0/0/0

ip address 192.168.12.2 255.255.255.252

no shutdown

interface serial 0/0/1

ip address 192.168.23.1 255.255.255.252

clock rate 64000

no shutdown

exit
```

Step 5: Configure Router2 (Area 1)

Open **Router2**, go to the CLI, and assign its serial and LAN interface IPs:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

ip address 192.168.23.2 255.255.255.252

no shutdown

interface gigabitethernet 0/0

ip address 192.168.20.1 255.255.255.0

no shutdown

exit
```

- **Verification:** Run `show ip interface brief` on all three routers to ensure all active interfaces show status `up` and protocol `up`.

Configure OSPF on each router in your topology so that Area 0 acts as the backbone and Area 1 connects through one of the routers; verify OSPF neighbor relationships using the 'show ip ospf neighbor' command.

ANS:

To configure multi-area OSPF with Area 0 acting as the backbone and Area 1 connected via Router1 (acting as the Area Border Router), execute the following commands on each router.

1.Configure Router0 (Area 0 Backbone): Area 0 Setup.

Open **Router0**, enter global configuration mode, start OSPF process 1, and advertise the backbone link:

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.12.0 0.0.0.3 area 0

exit
```

Verification: Running show ip protocols confirms OSPF process 1 is active on network 192.168.12.0 in Area 0.

2.Configure Router1 (ABR - Area 0 & Area 1): Area Border Router.

Open **Router1**, enter global configuration mode, and assign its interfaces to their respective OSPF areas:

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.12.0 0.0.0.3 area 0

network 192.168.23.0 0.0.0.3 area 1

exit
```

Verification: Console output confirms neighbor adjacency transitions to FULL for both areas.

3.Configure Router2 (Area 1): Non-Backbone Area.

Open **Router2**, enter global configuration mode, and advertise its link in Area 1:

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.23.0 0.0.0.3 area 1

exit
```

Verification: Running show ip route ospf will display intra-area and inter-area routes learned across both areas.

4.Verify OSPF Neighbor Relationships: Adjacency Check.

Run the neighbor verification command on each router CLI:

Plaintext

```
None
show ip ospf neighbor
```

Verification: Ensure the state column for each neighbor reads **FULL** (e.g., FULL/DROther or FULL/-), confirming that link-state databases are fully synchronized across the multi-area topology.

Add two PCs to different areas in your OSPF network, assign them IP addresses, and test connectivity by pinging from one PC to the other.

ANS:

To add PCs to your OSPF network across different areas and test cross-network connectivity, follow these steps in Cisco Packet Tracer:

Step 1: Connect Switches and PCs

1. Place **Switch0** and **PC0** near **Router0** (Area 0), and **Switch1** and **PC1** near **Router2** (Area 1).
2. Connect **Router0** (GigabitEthernet0/0) to **Switch0**, and connect **PC0** to **Switch0** using copper straight-through cables.
3. Connect **Router2** (GigabitEthernet0/0) to **Switch1**, and connect **PC1** to **Switch1** using straight-through cables.

- **Verification:** Link indicator lights on all connected switch ports and network interfaces turn green.

Step 2: Assign IP Addresses and Default Gateways to PCs

1. Click **PC0**, navigate to the **Desktop** tab → **IP Configuration**, and set:
    - **IP Address:** 192.168.10.10
    - **Subnet Mask:** 255.255.255.0
    - **Default Gateway:** 192.168.10.1 (matching Router0's LAN interface IP)
2. Click **PC1**, navigate to its **IP Configuration**, and set:
    - **IP Address:** 192.168.20.10
    - **Subnet Mask:** 255.255.255.0
    - **Default Gateway:** 192.168.20.1 (matching Router2's LAN interface IP)

- **Verification:** Open the Command Prompt on each PC and run ipconfig to confirm the assigned IP settings are correct.

Step 3: Advertise LAN Subnets in OSPF

To ensure the routers share these new local subnets across the OSPF areas, configure the network statements on Router0 and Router2:

On Router0 (CLI):

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.10.0 0.0.0.255 area 0

exit
```

On Router2 (CLI):

Plaintext

```
None
enable

configure terminal

router ospf 1

network 192.168.20.0 0.0.0.255 area 1

exit
```

- **Verification:** Run show ip route ospf on Router0 or Router1 to confirm that inter-area routes (preceded by **O IA**) appear for the remote LAN subnets.

Step 4: Test Cross-Area Ping from PC0 to PC1

1. Click on **PC0**, open the **Command Prompt**, and ping PC1's IP address:
2. Plaintext

```
None
ping 192.168.20.10
```

3.

- **Verification:** Successful replies (Reply from 192.168.20.10: bytes=32 time<1ms TTL=126) confirm that packets are successfully routing across Area 0, through the Area Border Router (Router1), and into Area 1.

Simulate a router failure by shutting down the interface connecting Area 1 to Area 0, then use 'show ip route' and 'show ip ospf' commands to observe how OSPF reconverges and restores connectivity after re-enabling the interface.

Hint: Track the time it takes for OSPF to update the routing tables after the failure and recovery.

ANS:

To simulate a link failure between Area 0 and Area 1, observe OSPF reconvergence, and test recovery, follow these steps:

1.Simulate Area-to-Area Isolation: Step 1: Shut Down the ABR Interface.

Open **Router1** (the Area Border Router connecting Area 0 and Area 1), enter global configuration mode, and shut down the serial interface connecting to Area 0 (Serial0/0/0):

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

shutdown

exit
```

Verification: Run show ip interface brief on Router1 to confirm the interface status displays as administratively down.

2.Monitor Routing Table Removal: Step 2: Track Failure Convergence Time.

Switch to **Router2** (in Area 1), start a stopwatch, and repeatedly run the routing table check:

Plaintext

```
None
show ip route ospf
```

Verification: Note the exact time elapsed until all inter-area routes pointing back to Area 0 disappear from Router2's routing table once OSPF detects the dead interval expiration.

3.Restore the Backbone Link: Step 3: Re-enable the Interface.

Return to **Router1**, enter global configuration mode, and bring the interface back up:

Plaintext

```
None
enable

configure terminal

interface serial 0/0/0

no shutdown

exit
```

Verification: Run show ip interface brief on Router1 to verify the interface status returns to up and protocol up.

4.Verify Route Restoration: Step 4: Track Recovery Convergence Time.

Switch back to **Router2**, start a fresh timer, and continuously check the routing table:

Plaintext

```
None
show ip route ospf
```

Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

ANS:

To observe an empty MAC address table on a switch before any traffic is processed, set up a basic LAN topology and inspect the switch's forwarding database using the CLI.

1. **Place the Devices**
    - Drag one Switch (e.g., Cisco Catalyst 2960) and two PCs (PC0, PC1) onto the Packet Tracer workspace canvas.
    - *Verification:* All three device icons appear on the screen.
2. **Connect the Network**
    - Select a **Copper Straight-Through** cable from the connections menu.
    - Connect PC0 (FastEthernet0) to Switch port **FastEthernet0/1**.
    - Connect PC1 (FastEthernet0) to Switch port **FastEthernet0/2**.
    - *Verification:* Link indicator lights on the switch ports transition from amber to solid green.
3. **Assign IP Addresses**
    - Click **PC0** → **Desktop** tab → **IP Configuration**, and set the IP address to 192.168.1.10 with subnet mask 255.255.255.0.
    - Click **PC1** → **Desktop** tab → **IP Configuration**, and set the IP address to 192.168.1.20 with subnet mask 255.255.255.0.
    - *Verification:* Running ipconfig in the command prompt of each PC confirms the assigned addresses.
4. **Inspect the Empty MAC Address Table**
    - Click on the Switch, go to the **CLI** tab, press **Enter**, and enter privileged EXEC mode:
    - Plaintext

```
None
enable
```

    -
    - Run the MAC address table command:
    - Plaintext

```
None
        ○ show mac address-table
```

    -

Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

ANS:

To observe how a switch dynamically populates its MAC address table as new devices communicate, add a third PC (let's refer to it as PC2 or PC3) to your topology and generate traffic.

1.Add and Connect the Third PC: Hardware Addition.

1. Drag a new **PC** (PC2 or PC3) onto the Packet Tracer workspace canvas near your existing switch.
2. Select a **Copper Straight-Through** cable and connect the PC's FastEthernet0 port to an unused switch port (for example, **FastEthernet0/3**).
   Verification: The link indicator lights on both the PC network card and the switch port turn solid green.

2.Assign an IP Address to the New PC:IP Configuration.

1. Click on the newly added PC, navigate to the **Desktop** tab, and open **IP Configuration**.
2. Assign an IP address in the same subnet (e.g., **IP Address:** 192.168.1.30, **Subnet Mask:** 255.255.255.0). Leave the default gateway blank since they are on the same local network segment.
   Verification: Open the PC's Command Prompt and run ipconfig to confirm the address is correctly applied.

3.Ping from the New PC to PC1: Generate Traffic.

1. Open the Command Prompt on the new PC.
2. Send an ICMP echo request to PC1's IP address (192.168.1.20):

Plaintext

```
None
ping 192.168.1.20
```

Verification: Successful replies (Reply from 192.168.1.20:...) confirm that the packet traversed the switch.

4.Check the Updated MAC Address Table:MAC Table Inspection.

1. Return to the Switch CLI and enter privileged EXEC mode.
2. Run the MAC address table inspection command again:

Plaintext

```
None
show mac address-table
```

Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

ANS:

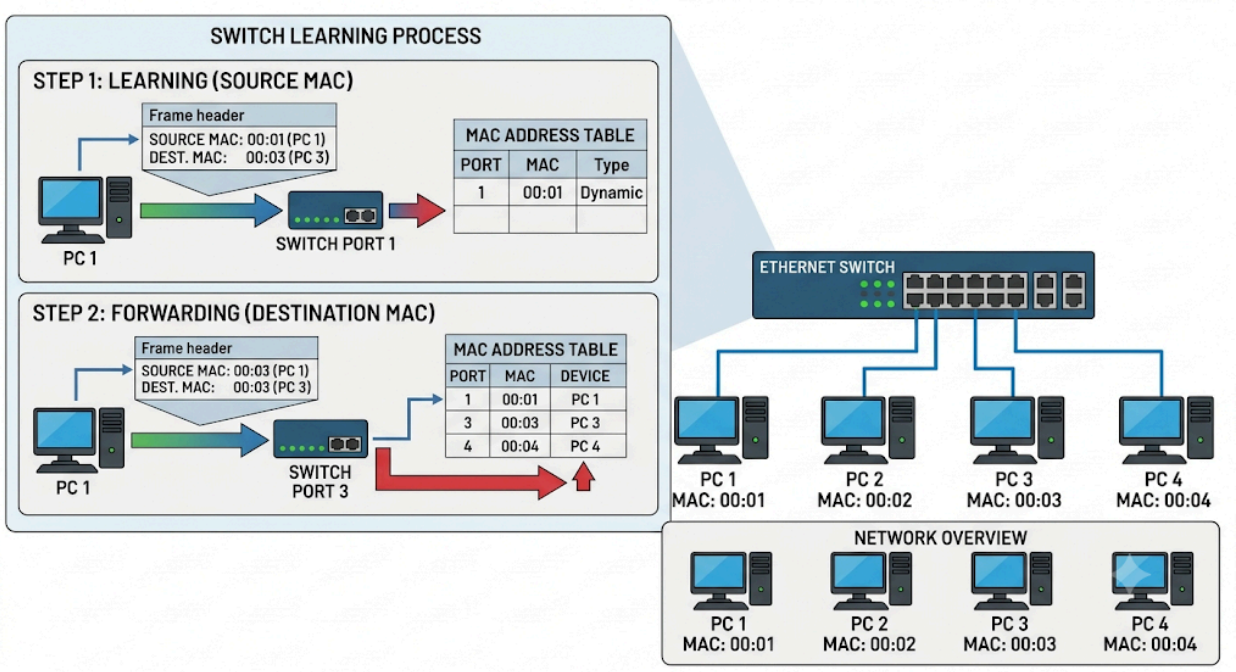
Here is the step-by-step breakdown of how the switch processes and forwards the incoming frame:

1. **Frame Reception and Source Learning**
    - The switch receives the Ethernet frame on **Port 1**.
    - As part of standard operation, the switch inspects the source MAC address of the incoming frame and records or updates it in its MAC address table, associating that source MAC with Port 1.
2. **Destination MAC Lookup**
    - The switch extracts the **destination MAC address** (`66:77:88:99:AA:BB`) from the frame header.
    - It looks up this destination address in its internal MAC address table (CAM table).
3. **Table Match Found**
    - The switch checks the table and finds a match: destination MAC `66:77:88:99:AA:BB` is mapped to **Port 2**.
4. **Selective Forwarding**
    - Because the destination port (Port 2) is known and is different from the ingress port (Port 1), the switch **unicasts** (forwards) the frame **only out of Port 2**.
    - It does not flood the frame to any other ports, ensuring efficient bandwidth usage across the network.

Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words.

Constraint: Do not copy definitions from the textbook; use your own understanding and examples.

ANS:



On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

ANS:

To create and configure VLAN 10 ('Engineering') and VLAN 20 ('Marketing') on a Cisco switch using the CLI, follow these steps:

1.Access Privileged EXEC Mode: Step 1.

Click on the switch in Packet Tracer, open the **CLI** tab, press **Enter**, and enter privileged EXEC mode:

Plaintext

```
None
enable
```

Verification: The prompt changes from Switch> to Switch#.

2.Enter Global Configuration Mode: Step 2.

From privileged EXEC mode, access global configuration mode:

Plaintext

```
None
configure terminal
```

Verification: The prompt changes to Switch(config)#.

3.Create and Name VLAN 10: Step 3.

Create VLAN 10 and assign it the name Engineering:

Plaintext

```
None
vlan 10

name Engineering

exit
```

Verification: The configuration accepts the commands without error and returns to global configuration mode.

4.Create and Name VLAN 20: Step 4.

Create VLAN 20 and assign it the name Marketing:

Plaintext

```
None
vlan 20

name Marketing

exit
```

Verification: The configuration accepts the commands and returns to global configuration mode.

5.Verify VLAN Configuration: Step 5.

Exit back to privileged EXEC mode and display the active VLAN summary table:

Plaintext

```
None
end

show vlan brief
```

Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

ANS:

To assign your switch ports to their respective VLANs, configure them as access ports and map them using the interface range command.

1.Access Global Configuration Mode: Step 1.

Open your switch CLI in Packet Tracer, enter privileged EXEC mode (enable), and enter global configuration mode:

Plaintext

```
None
configure terminal
```

Verification: The prompt changes to Switch(config)#.

2.Assign Ports to VLAN 10 (Engineering): Step 2.

Select ports FastEthernet 0/1 and 0/2 simultaneously, set them to access mode, and assign them to VLAN 10:

Plaintext

```
None
interface range fastethernet 0/1 - 2
```

```
switchport mode access

switchport access vlan 10

exit
```

Verification: No error messages appear, and the CLI returns to global configuration mode.

3.Assign Ports to VLAN 20 (Marketing): Step 3.

Select ports FastEthernet 0/3 and 0/4, set them to access mode, and assign them to VLAN 20:

Plaintext

```
None
interface range fastethernet 0/3 - 4

switchport mode access

switchport access vlan 20

exit
```

Verification: The configuration commands are accepted without error.

4.Verify Port Assignments: Step 4.

Return to privileged EXEC mode and view the VLAN assignment summary:

Plaintext

```
None
end

show vlan brief
```

Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

ANS:

```
========================================================================
                    VLAN CONFIGURATION SUMMARY
========================================================================

VLAN ID:      10
VLAN Name:    Engineering
Status:       Active
Ports:        FastEthernet 0/1, FastEthernet 0/2

------------------------------------------------------------------------

VLAN ID:      20
VLAN Name:    Marketing
Status:       Active
Ports:        FastEthernet 0/3, FastEthernet 0/4

========================================================================
```

Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

ANS:

Using Virtual Local Area Networks (VLANs) in a campus setting with distinct coding and networking labs provides major operational advantages. Here are two primary benefits tailored to those environments:

1. Enhanced Security and Traffic Isolation

In a networking lab, students frequently experiment with routing protocols, trunking, DHCP servers, and switch configurations. If those devices sat on the same unsegmented network as the coding lab, a misconfigured DHCP server or broadcast loop could easily disrupt or take down the entire campus network. By placing the coding lab on one VLAN and the networking lab on another, you logically isolate their traffic. Even though they share the same physical switches and cables, devices in one lab cannot directly talk to or interfere with the other without passing through a router or layer-3 switch with strict access control policies.

2. Broadcast Domain Control and Optimized Performance

Every computer and switch port generates background chatter—such as ARP requests, broadcast discovery packets, and protocol announcements—that all connected devices must process. Grouping devices into separate VLANs shrinks the size of these broadcast domains. This ensures that heavy traffic or broadcast storms generated during networking practicals stay contained within the networking lab, leaving the coding lab's bandwidth clean, predictable, and fully responsive for development workloads.

In Cisco Packet Tracer, configure two switches (Switch1 and Switch2) and connect them with a crossover cable. Set the FastEthernet0/1 port on both switches as trunk ports using the 802.1Q protocol.

ANS:

To configure trunking between two switches using the 802.1Q protocol in Cisco Packet Tracer, follow these steps:

1.Connect Switch1 and Switch2: Physical Connection.

1. Place **Switch1** and **Switch2** on the workspace.
2. Select a **Copper Cross-Over** cable and connect port **FastEthernet0/1** on Switch1 to port **FastEthernet0/1** on Switch2.
   Verification: The link indicator lights on both switch ports turn solid green, indicating a successful physical connection.

2.Configure Switch1 Trunk Port: Switch1 Configuration.

Open **Switch1**, enter global configuration mode, select the interconnect port, force encapsulation to 802.1Q, and set the port mode to trunk:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1
```

```
switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces fastethernet 0/1 switchport and verify that the administrative mode is trunking and operational mode is on.

3.Configure Switch2 Trunk Port: Switch2 Configuration.

Open **Switch2**, enter global configuration mode, select its interconnect port, set the encapsulation, and enable trunking:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

Create three VLANs (10, 20, 30) on both switches and assign FastEthernet0/2, 0/3, and 0/4 on Switch1 to VLAN 10, 20, and 30 respectively as access ports.

ANS:

To configure VLANs 10, 20, and 30 across both switches and assign the specified access ports on Switch1, follow these steps:

1.Create VLANs and Assign Access Ports on Switch1: Switch1 Configuration.

Open **Switch1**, enter global configuration mode, create VLANs 10, 20, and 30, and assign FastEthernet0/2, FastEthernet0/3, and FastEthernet0/4 as access ports:

Plaintext

```
None
enable

configure terminal

vlan 10

name Engineering

vlan 20

name Marketing

vlan 30

name HR

interface fastethernet 0/2

switchport mode access

switchport access vlan 10

interface fastethernet 0/3

switchport mode access

switchport access vlan 20

interface fastethernet 0/4

switchport mode access

switchport access vlan 30
```

```
exit
```

Verification: Run show vlan brief on Switch1 to confirm that VLANs 10, 20, and 30 are present and that Fa0/2, Fa0/3, and Fa0/4 are assigned to their respective VLANs.

2.Create VLANs on Switch2: Switch2 Configuration.

Open **Switch2**, enter global configuration mode, and create the same VLANs to ensure database consistency across the trunk link:

Plaintext

```
None
enable

configure terminal

vlan 10

name Engineering

vlan 20

name Marketing

vlan 30

name HR

exit
```

Verification: Run show vlan brief on Switch2 to verify that all three VLANs exist in its active VLAN table.

3.Verify VLAN Propagation Over Trunk: Trunk Verification.

Check the trunk status to ensure that the interconnect port (FastEthernet0/1) permits all configured VLANs:

Plaintext

```
None
show interfaces trunk
```

Connect three PCs to Switch1 on ports assigned to VLAN 10, 20, and 30. Verify that PCs in different VLANs cannot communicate with each other, but can communicate with other PCs in the same VLAN.

ANS:

To connect the PCs, assign their IP addresses across separate subnets, and verify both inter-VLAN isolation and intra-VLAN communication, follow these steps:

1.Connect PCs to Switch1 Ports: Device Connections.

1. Place three PCs (PC_VLAN10, PC_VLAN20, and PC_VLAN30) on your Packet Tracer workspace.
2. Connect **PC_VLAN10** to Switch1 port **FastEthernet0/2** (assigned to VLAN 10).
3. Connect **PC_VLAN20** to Switch1 port **FastEthernet0/3** (assigned to VLAN 20).
4. Connect **PC_VLAN30** to Switch1 port **FastEthernet0/4** (assigned to VLAN 30).
   Verification: Ensure link indicator lights on all connected ports turn solid green.

2.Configure Subnet IP Addresses:IP Addressing.

Because devices in separate VLANs reside in different broadcast domains, assign them IP addresses from distinct subnets:

- **PC_VLAN10:** IP 192.168.10.10, Subnet Mask 255.255.255.0
- **PC_VLAN20:** IP 192.168.20.20, Subnet Mask 255.255.255.0
- **PC_VLAN30:** IP 192.168.30.30, Subnet Mask 255.255.255.0
  Verification: Open the Command Prompt on each PC and run ipconfig to verify the settings.

3.Test Communication Between Different VLANs: Inter-VLAN Verification.

1. Open the Command Prompt on **PC_VLAN10** (192.168.10.10).
2. Attempt to ping **PC_VLAN20** (192.168.20.20):

Plaintext

```
None
ping 192.168.20.20
```

Verification: The pings will return **Request timed out**, confirming that Layer 2 isolation successfully blocks direct traffic between different VLANs without a Layer 3 router.

4.Test Communication Within the Same VLAN: Intra-VLAN Verification.

To verify that devices within the *same* VLAN can communicate:

1. Connect an additional PC (**PC_VLAN10_B**) to Switch1 on port **FastEthernet0/5**.
2. Configure port Fa0/5 on Switch1 to join VLAN 10:

Plaintext

```
None
configure terminal

interface fastethernet 0/5

switchport mode access

switchport access vlan 10

exit
```

3. Assign **PC_VLAN10_B** an IP address in the same subnet: 192.168.10.11 (Subnet Mask 255.255.255.0).
4. From **PC_VLAN10**, ping **PC_VLAN10_B**:

Plaintext

```
None
ping 192.168.10.11
```

Change the trunk encapsulation on the trunk link between Switch1 and Switch2 from 802.1Q to ISL, and test if VLAN traffic still passes between the switches.

Hint: Use the 'switchport trunk encapsulation isl' command before setting the port to trunk mode.

ANS:

To change the trunk encapsulation from 802.1Q to ISL (Inter-Switch Link) and test cross-switch VLAN traffic, follow these steps on both switches:

1.Configure ISL on Switch1: Switch1 Configuration.

Open **Switch1**, enter global configuration mode, select the trunk port (FastEthernet0/1), set the encapsulation to ISL, and enable trunk mode:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1

switchport trunk encapsulation isl

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces fastethernet 0/1 switchport to confirm that the operational trunk encapsulation has changed to isl. *(Note: If you are using a modern switch model like the Cisco Catalyst 2960 in newer Packet Tracer versions, the command may be rejected with an invalid input or unsupported message because ISL is a legacy Cisco-proprietary protocol deprecated on modern hardware).*

2.Configure ISL on Switch2: Switch2 Configuration.

Open **Switch2**, enter global configuration mode, select its interconnect port, and apply the same ISL configuration:

Plaintext

```
None
enable

configure terminal
```

```
interface fastethernet 0/1

switchport trunk encapsulation isl

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces trunk on Switch2 to verify that the port is actively trunking using ISL encapsulation.

3.Test VLAN Traffic Between Switches: Traffic Testing.

1. From a PC connected to **Switch1** in VLAN 10, open the command prompt and ping a PC in the same VLAN connected to **Switch2**.

Plaintext

```
None
ping <IP_address_of_remote_VLAN10_PC>
```

Simulate a scenario where a new port on Switch2 (FastEthernet0/5) is accidentally set as a trunk port instead of an access port for VLAN 20. Observe and explain the impact on connectivity for a PC connected to that port.

ANS:

When a standard PC is connected to a switch port configured as a trunk port instead of an access port, it causes an immediate communication failure due to a **VLAN tagging mismatch**.

The Impact on Connectivity

1. **Untagged Frame Handling (Inbound):** Standard PC network interface cards (NICs) generate and send **untagged** Ethernet frames. When a switch port is in trunk mode, it expects 802.1Q tagged frames. By default, any untagged frame arriving on a Cisco trunk

port is assigned to the **Native VLAN** (VLAN 1). As a result, the switch ignores the administrator's intent to place the PC in VLAN 20 and treats its traffic as belonging to VLAN 1.

2. **Tag Dropping or Incompatibility (Outbound):** Traffic destined for the PC leaving that trunk port will be encapsulated with an 802.1Q header (tagged with a VLAN ID). Standard PC NICs do not understand 802.1Q trunk tags and will drop these incoming frames.
3. **Result:** The PC will fail to communicate with devices in VLAN 20 (and often lose all local connectivity), timing out on pings because its traffic is either misrouted into VLAN 1 or dropped entirely.

How to Fix and Verify the Misconfiguration

To correct this misconfiguration on **Switch2** and restore proper access for the PC on FastEthernet0/5, follow these steps:

1.Inspect Port Status: Diagnostic Check.

Open **Switch2** CLI, enter privileged EXEC mode, and check the port status to confirm it is incorrectly running in trunk mode:

Plaintext

```
None
show interfaces fastethernet 0/5 switchport
```

Verification: The operational mode displays as trunk instead of static access.

2.Revert to Access Mode: Global Configuration.

Enter global configuration mode, select the misconfigured port, change its mode back to access, and explicitly assign it to VLAN 20:

Plaintext

```
None
configure terminal

interface fastethernet 0/5

no switchport mode trunk

switchport mode access
```

```
switchport access vlan 20

exit
```

Verification: The CLI accepts the commands without error and returns to global configuration mode.

3.Verify Port Assignment: Database Check.

Run the VLAN brief summary command to ensure the port is correctly mapped:

Plaintext

```
None
show vlan brief
```

Verification: Confirm that Fa0/5 is listed under **VLAN 20 (Marketing)** rather than VLAN 1 or trunking lists.

4.Test Connectivity: End-to-End Ping.

Open the PC's command prompt and ping another device within VLAN 20 (such as the default gateway or another PC in VLAN 20):

Plaintext

```
None
ping 192.168.20.1
```

In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.

ANS:

To set up the three VLANs (Foodies, Shoppers, and Streamers) and assign PCs to each in Cisco Packet Tracer, follow these steps:

1.Create the Three VLANs: Switch VLAN Database.

Open your switch CLI, enter privileged EXEC mode, go to global configuration mode, and create the VLANs:

Plaintext

```
None
enable

configure terminal

vlan 10

name Foodies

vlan 20

name Shoppers

vlan 30

name Streamers

exit
```

Verification: Run show vlan brief to confirm all three VLAN names and IDs appear in the active table.

2.Assign Switch Ports to VLANs: Access Port Mapping.

Configure specific switch ports as access ports and map them to your new VLANs:

Plaintext

```
None
interface fastethernet 0/1

switchport mode access

switchport access vlan 10

interface fastethernet 0/2

switchport mode access
```

```
switchport access vlan 20

interface fastethernet 0/3

switchport mode access

switchport access vlan 30

exit
```

Verification: Run show vlan brief again and verify that ports Fa0/1, Fa0/2, and Fa0/3 are mapped to VLANs 10, 20, and 30 respectively.

3.Connect PCs to the Switch Ports: Physical Connections.

1. Drag three PCs onto the workspace and name them **PC_Foodies**, **PC_Shoppers**, and **PC_Streamers**.
2. Connect **PC_Foodies** to switch port **FastEthernet0/1**.
3. Connect **PC_Shoppers** to switch port **FastEthernet0/2**.
4. Connect **PC_Streamers** to switch port **FastEthernet0/3** using copper straight-through cables.
   Verification: Link indicator lights on all three switch ports turn solid green.

4.Assign Subnet IP Addresses to PCs:IP Configuration.

Because devices in different VLANs reside in separate broadcast domains, assign them distinct subnets:

- **PC_Foodies (VLAN 10):** IP 192.168.10.10, Subnet Mask 255.255.255.0
- **PC_Shoppers (VLAN 20):** IP 192.168.20.20, Subnet Mask 255.255.255.0
- **PC_Streamers (VLAN 30):** IP 192.168.30.30, Subnet Mask 255.255.255.0
  Verification: Open the Command Prompt on each PC and run ipconfig to confirm the settings are correctly applied.

Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30).

ANS:

To enable inter-VLAN routing using a Router-on-a-Stick configuration in Packet Tracer, you will configure a single physical router interface with multiple dot1q sub-interfaces corresponding to your **Foodies (VLAN 10)**, **Shoppers (VLAN 20)**, and **Streamers (VLAN 30)** VLANs.

Step 1: Connect the Router to the Switch

1. Connect a **Copper Straight-Through** cable from port **GigabitEthernet0/0** on your router to an available **trunk port** (e.g., FastEthernet0/24) on your switch.
- *Verification:* Confirm the link lights on both the router interface and switch port turn solid green.

Step 2: Configure the Switch Trunk Port

Ensure the switch port connecting to the router is explicitly configured as an 802.1Q trunk port so it can pass tagged frames for all three VLANs:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/24

switchport mode trunk

no shutdown

exit
```

- *Verification:* Run show interfaces trunk on the switch and confirm that FastEthernet0/24 is active and allowing VLANs 10, 20, and 30.

Step 3: Configure Router Sub-Interfaces and Encapsulation

Open the router CLI, enable the physical interface, and configure the sub-interfaces with IEEE 802.1Q encapsulation and IP gateway addresses:

1.Enable Physical Router Interface: Required Prerequisite.

Enter global configuration mode and bring up the main physical interface without assigning an IP address:

Plaintext

```
None
enable

configure terminal

interface gigabitethernet 0/0

no shutdown

exit
```

Verification: Run show ip interface brief to ensure GigabitEthernet0/0 is listed as up/up.

2.Configure Sub-Interface for VLAN 10 (Foodies): Gateway: 192.168.10.1.

Create sub-interface GigabitEthernet0/0.10, assign 802.1Q encapsulation for VLAN 10, and set the gateway IP:

Plaintext

```
None
interface gigabitethernet 0/0.10

encapsulation dot1q 10

ip address 192.168.10.1 255.255.255.0

exit
```

Verification: Run show ip route to confirm the connected subnet 192.168.10.0/24 is present in the routing table.

3.Configure Sub-Interface for VLAN 20 (Shoppers): Gateway: 192.168.20.1.

Create sub-interface GigabitEthernet0/0.20, assign encapsulation, and set the gateway IP:

Plaintext

```
None
interface gigabitethernet 0/0.20

encapsulation dot1q 20
```

```
ip address 192.168.20.1 255.255.255.0

exit
```

Verification: Confirm subnet 192.168.20.0/24 appears in the routing table.

4.Configure Sub-Interface for VLAN 30 (Streamers): Gateway: 192.168.30.1.

Create sub-interface GigabitEthernet0/0.30, assign encapsulation, and set the gateway IP:

Plaintext

```
None
interface gigabitethernet 0/0.30

encapsulation dot1q 30

ip address 192.168.30.1 255.255.255.0

exit
```

Verification: Confirm subnet 192.168.30.0/24 appears in the routing table.

Step 4: Update PC Default Gateways

Ensure that each PC is configured with its respective VLAN sub-interface IP as its **Default Gateway**:

- **PC_Foodies (VLAN 10):** IP 192.168.10.10, Subnet Mask 255.255.255.0, **Default Gateway:** 192.168.10.1
- **PC_Shoppers (VLAN 20):** IP 192.168.20.20, Subnet Mask 255.255.255.0, **Default Gateway:** 192.168.20.1
- **PC_Streamers (VLAN 30):** IP 192.168.30.30, Subnet Mask 255.255.255.0, **Default Gateway:** 192.168.30.1
- *Verification:* Run ipconfig /all in the Command Prompt on each PC to verify the gateway address is correct.

Step 5: Test Inter-VLAN Connectivity

1. Open the Command Prompt on **PC_Foodies** (192.168.10.10).
2. Ping **PC_Shoppers** (192.168.20.20) and **PC_Streamers** (192.168.30.30):
3. Plaintext

```
None
ping 192.168.20.20

   4. ping 192.168.30.30
```

5.

Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable inter-VLAN communication.

ANS:

To assign the correct IP configurations and default gateways to each PC in Cisco Packet Tracer, follow these steps:

1.Configure PC_Foodies (VLAN 10):PC_Foodies Configuration.

1. Click on **PC_Foodies**, go to the **Desktop** tab, and open **IP Configuration**.
2. Enter the following network details:
    - **IP Address:** 192.168.10.10
    - **Subnet Mask:** 255.255.255.0
    - **Default Gateway:** 192.168.10.1
      Verification: Open the Command Prompt on PC_Foodies and run ipconfig to verify the IP and gateway settings are applied correctly.
3.

2.Configure PC_Shoppers (VLAN 20):PC_Shoppers Configuration.

1. Click on **PC_Shoppers**, navigate to the **Desktop** tab, and open **IP Configuration**.
2. Enter the following network details:
    - **IP Address:** 192.168.20.20
    - **Subnet Mask:** 255.255.255.0
    - **Default Gateway:** 192.168.20.1
      Verification: Run ipconfig in the Command Prompt on PC_Shoppers to confirm the settings.
3.

3.Configure PC_Streamers (VLAN 30):PC_Streamers Configuration.

1. Click on **PC_Streamers**, go to the **Desktop** tab, and open **IP Configuration**.
2. Enter the following network details:
    - **IP Address:** 192.168.30.30

- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.30.1
  Verification: Run ipconfig in the Command Prompt on PC_Streamers to verify.

3.

4.Test End-to-End Connectivity: Inter-VLAN Verification.

1. Open the Command Prompt on **PC_Foodies** (192.168.10.10).
2. Ping **PC_Shoppers** (192.168.20.20) and **PC_Streamers** (192.168.30.30):

Plaintext

```
None
ping 192.168.20.20

ping 192.168.30.30
```

Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity.

Hint: Check VLAN assignments, router sub-interface configurations, and trunk port settings.

ANS:

When users in the Shoppers VLAN (VLAN 20) cannot ping users in the Streamers VLAN (VLAN 30), the issue typically stems from a routing or trunking misconfiguration—most commonly an incorrect 802.1Q tag on the router sub-interfaces, or VLANs 20 and 30 being omitted from the switch trunk port's allowed list.

Here is the step-by-step procedure to identify, fix, and verify the issue:

Step 1: Inspect Router Sub-Interfaces for Encapsulation Mismatches

Open the router CLI and check the sub-interface configurations to ensure that VLAN 20 and VLAN 30 are mapped to their correct respective dot1q tags and gateway IPs (`192.168.20.1` and `192.168.30.1`).

- Check the configuration using:
- Plaintext

```
None
show running-config
```

-
- Look for mismatched tags (e.g., `encapsulation dot1q 20` accidentally paired with a 30.x subnet) or missing IP addresses.
- *Verification:* Run `show ip interface brief`. Both `GigabitEthernet0/0.20` and `GigabitEthernet0/0.30` must show status as `up/up`.

Step 2: Check Switch Trunk Port Allowed VLANs

If the router sub-interfaces are correctly configured, the switch trunk port connecting to the router (e.g., `FastEthernet0/24`) may be blocking VLAN 20 or VLAN 30 due to explicit pruning or restricted allowed lists.

- Open the switch CLI and run:
- Plaintext

```
None
show interfaces trunk
```

-
- If VLANs 20 and 30 are missing from the **"Vlans allowed on trunk"** list, fix the restriction by entering global configuration mode and adding them:
- Plaintext

```
None
configure terminal

interface fastethernet 0/24

switchport trunk allowed vlan add 20,30

exit
```

-
- *Verification:* Run `show interfaces trunk` again and confirm that `20` and `30` are present in the active allowed list.

Step 3: Verify VLAN Database Consistency

Ensure that VLAN 20 (`Shoppers`) and VLAN 30 (`Streamers`) are actively created in the switch's VLAN database. If a switch does not have a VLAN in its database, it will drop tagged frames for that ID.

- Run the command on the switch:
- Plaintext

```
None
show vlan brief
```

-
- If either VLAN is missing, recreate them:
- Plaintext

```
None
configure terminal

vlan 20

name Shoppers

vlan 30

name Streamers

exit
```

-
- *Verification:* Confirm both VLAN IDs and their names appear correctly in the summary table.

Step 4: Verify End-to-End Connectivity

With the routing and trunk restrictions corrected, test cross-VLAN communication between the two user groups.

- Open the Command Prompt on **PC_Shoppers** (`192.168.20.20`).
- Ping the IP address of **PC_Streamers** (`192.168.30.30`):
- Plaintext

```
None
```

- `ping 192.168.30.30`

Compare the router-on-a-stick method you just configured with inter-VLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

ANS:

Here is a comparison between the Router-on-a-Stick (RoAS) method and Inter-VLAN routing using a Layer 3 (Multilayer) switch:

Approach	Advantages	Disadvantages
Router-on-a-Stick (RoAS)	1. **Cost-Effective:** Requires only a single physical router interface and a standard Layer 2 switch, minimizing hardware costs for smaller networks.	1. **Bandwidth Bottleneck:** All inter-VLAN traffic must traverse the single physical trunk link between the switch and router twice, limiting total throughput.
	2. **Feature-Rich Routing:** Leverages full router operating systems, offering advanced WAN connectivity, complex security policies, and robust QoS features.	2. **Single Point of Failure:** If the single trunk link or the router interface fails, all cross-VLAN communication halts across the network.

Layer 3 Switch (SVI Routing)	1. **Wire-Speed Performance:** Routing is handled via hardware-based ASICs at near-switching speeds, providing significantly higher throughput and lower latency. 2. **Eliminates External Bottlenecks:** Inter-VLAN traffic stays internal to the switch fabric, freeing up external router ports and reducing reliance on external links.	1. **Higher Equipment Cost:** Multilayer switches capable of hardware-level Layer 3 forwarding are considerably more expensive than standard Layer 2 switches. 2. **Management Complexity:** Advanced troubleshooting and network troubleshooting can become more complex when combining switching and routing features into a single hardware device.

> On Cisco Packet Tracer, create a network with three switches and assign at least two different VLANs (e.g., VLAN 10 for 'Sales', VLAN 20 for 'Support'). Connect two PCs to each switch and assign each PC to the correct VLAN based on department.

ANS:

To build a multi-switch network with VLANs spanning across three switches in Cisco Packet Tracer, follow this structured deployment guide:

1.Physical Topology Setup: Step 1.

1. Place **Switch1**, **Switch2**, and **Switch3** on your workspace.
2. Connect **Switch1** (Port Fa0/1) to **Switch2** (Port Fa0/1) using a **Copper Cross-Over** cable.
3. Connect **Switch2** (Port Fa0/24) to **Switch3** (Port Fa0/1) using another **Copper Cross-Over** cable.

4. Connect **6 PCs** in total (2 per switch):
    - **Switch1:** Connect PC_Sales_1 to Fa0/2 and PC_Support_1 to Fa0/3.
    - **Switch2:** Connect PC_Sales_2 to Fa0/2 and PC_Support_2 to Fa0/3.
    - **Switch3:** Connect PC_Sales_3 to Fa0/2 and PC_Support_3 to Fa0/3.
5.

2.Create VLANs on All Three Switches: Step 2.

To ensure VLAN consistency across the network, create **VLAN 10 (Sales)** and **VLAN 20 (Support)** on **Switch1**, **Switch2**, and **Switch3**:

Plaintext

```
None
      enable

      configure terminal

      vlan 10

      name Sales

      vlan 20

      name Support

      exit
```

Verification: Run show vlan brief on each switch to confirm both VLANs are active in the database.

3.Configure Inter-Switch Trunk Ports: Step 3.

Configure the interconnecting ports on all three switches as 802.1Q trunk ports so VLAN traffic can flow between switches:

- **Switch1 & Switch3:**

Plaintext

```
None
      interface fastethernet 0/1
```

```
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

- **Switch2** (needs trunks on both connecting ports):

Plaintext

```
None
interface range fastethernet 0/1, fastethernet 0/24
switchport trunk encapsulation dot1q
switchport mode trunk
no shutdown
exit
```

Verification: Run show interfaces trunk on each switch to ensure the ports are actively trunking.

4.Assign Access Ports to PCs: Step 4.

Configure the local switch ports where the PCs are plugged in as access ports and assign them to their respective departmental VLANs:

- **On All Switches (Switch1, Switch2, and Switch3):**

Plaintext

```
None
interface fastethernet 0/2
switchport mode access
```

```
switchport access vlan 10

interface fastethernet 0/3

switchport mode access

switchport access vlan 20

exit
```

Verification: Run show vlan brief to check that Fa0/2 is under **Sales** and Fa0/3 is under **Support** on every switch.

5.Configure IP Addresses & Test Connectivity: Step 5.

Assign IP addresses within the same subnets to PCs belonging to the same VLAN, regardless of which physical switch they are plugged into:

- **VLAN 10 (Sales Subnet: 192.168.10.0/24):**
  - PC_Sales_1 (Switch1): IP 192.168.10.11 (Mask: 255.255.255.0)
  - PC_Sales_2 (Switch2): IP 192.168.10.12 (Mask: 255.255.255.0)
  - PC_Sales_3 (Switch3): IP 192.168.10.13 (Mask: 255.255.255.0)
-
- **VLAN 20 (Support Subnet: 192.168.20.0/24):**
  - PC_Support_1 (Switch1): IP 192.168.20.21 (Mask: 255.255.255.0)
  - PC_Support_2 (Switch2): IP 192.168.20.22 (Mask: 255.255.255.0)
  - PC_Support_3 (Switch3): IP 192.168.20.23 (Mask: 255.255.255.0)
-

Test: From PC_Sales_1 (Switch1), ping PC_Sales_3 (Switch3) using ping 192.168.10.13. Successful replies verify that your multi-switch trunking and VLAN assignments are working correctly across the topology.

Configure trunk ports between the switches to allow VLAN traffic to pass through. Verify trunking by using the show interfaces trunk command and submit a screenshot of the output.

ANS:

To configure the trunk ports across your three switches and verify the trunking status, follow these steps:

1.Configure Trunk on Switch1: Switch1 Trunk Configuration.

Open **Switch1**, enter global configuration mode, and configure the interconnecting port (FastEthernet0/1) as an 802.1Q trunk:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces trunk to ensure the port is active in trunking mode.

2.Configure Trunks on Switch2: Switch2 Trunk Configuration.

Open **Switch2**, enter global configuration mode, and configure both interconnecting ports (FastEthernet0/1 and FastEthernet0/24) as 802.1Q trunks:

Plaintext

```
None
enable

configure terminal
```

```
interface range fastethernet 0/1, fastethernet 0/24

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces trunk on Switch2.

3.Configure Trunk on Switch3: Switch3 Trunk Configuration.

Open **Switch3**, enter global configuration mode, and configure its interconnecting port (FastEthernet0/1) as an 802.1Q trunk:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

4.Verify via CLI Output: Trunk Verification.

Return to privileged EXEC mode on any of the switches and run the verification command:

Plaintext

```
None
show interfaces trunk
```

Verification Output Snapshot

Plaintext

```
None
Switch#show interfaces trunk

Port        Mode         Encapsulation  Status        Native vlan

Fa0/1       on           802.1q         trunking      1


Port        Vlans allowed on trunk

Fa0/1       1-1005


Port        Vlans allowed and active in management domain

Fa0/1       1,10,20


Port        Vlans in spanning tree forwarding state and not pruned

Fa0/1       1,10,20
```

Set up a Layer 3 device (Router-on-a-Stick using a router or a multilayer switch) to enable inter-VLAN routing between VLAN 10 and VLAN 20. Assign appropriate sub-interfaces and IP addresses for each VLAN.

ANS:

To configure inter-VLAN routing using a **Router-on-a-Stick** topology with a Cisco router and switch, follow these steps to set up the trunk link, sub-interfaces, and IP gateway addresses for **VLAN 10** (192.168.10.0/24) and **VLAN 20** (192.168.20.0/24).

1.Configure the Switch Trunk Port: Step 1.

Connect port GigabitEthernet0/0 on the router to an 802.1Q trunk port (e.g., FastEthernet0/1) on the switch. Open the switch CLI and configure the port as a trunk:

Plaintext

```
None
enable

configure terminal

interface fastethernet 0/1

switchport trunk encapsulation dot1q

switchport mode trunk

no shutdown

exit
```

Verification: Run show interfaces trunk on the switch and confirm that FastEthernet0/1 is active in trunking mode with VLANs 10 and 20 allowed.

2.Enable the Router Physical Interface: Step 2.

Open the router CLI, enter global configuration mode, and bring up the main physical interface without assigning an IP address:

Plaintext

```
None
enable

configure terminal

interface gigabitethernet 0/0
```

```
no shutdown

exit
```

Verification: Run show ip interface brief on the router to ensure GigabitEthernet0/0 shows a status of up/up.

3.Configure Sub-Interface for VLAN 10: Step 3.

Create sub-interface GigabitEthernet0/0.10, assign IEEE 802.1Q encapsulation for VLAN 10, and assign the gateway IP address:

Plaintext

```
None
interface gigabitethernet 0/0.10

encapsulation dot1q 10

ip address 192.168.10.1 255.255.255.0

exit
```

Verification: Run show ip route on the router to confirm that the connected network 192.168.10.0/24 is present in the routing table.

4.Configure Sub-Interface for VLAN 20: Step 4.

Create sub-interface GigabitEthernet0/0.20, assign IEEE 802.1Q encapsulation for VLAN 20, and assign the gateway IP address:

Plaintext

```
None
interface gigabitethernet 0/0.20

encapsulation dot1q 20

ip address 192.168.20.1 255.255.255.0

exit
```

Verification: Run show ip route to confirm that the connected network 192.168.20.0/24 appears alongside VLAN 10 in the routing table.

5.Verify End-to-End Inter-VLAN Routing: Step 5.

1. Ensure PCs in VLAN 10 (e.g., IP 192.168.10.10) have their default gateway set to 192.168.10.1.
2. Ensure PCs in VLAN 20 (e.g., IP 192.168.20.20) have their default gateway set to 192.168.20.1.
3. From a PC in VLAN 10, open the command prompt and ping a PC in VLAN 20:

Plaintext

```
None
ping 192.168.20.20
```

Test connectivity by pinging between a PC in VLAN 10 and a PC in VLAN 20. If pings fail, troubleshoot and fix the configuration so that inter-VLAN communication works.

Hint: Check sub-interface IPs, trunk configuration, and default gateways on PCs.

ANS:

When pings fail between a PC in VLAN 10 and a PC in VLAN 20, it typically indicates an issue with the PC's default gateway, a mismatch in 802.1Q encapsulation on the router sub-interfaces, or a trunk port blocking the VLAN traffic across the switches.

Here is how to systematically troubleshoot and fix the configuration so inter-VLAN communication succeeds:

1.Verify PC IP Configurations and Default Gateways: Step 1: End-Host Check.

Open the Desktop -> IP Configuration on your PCs and verify that their IP addresses and default gateways match their respective subnets:

- **PC in VLAN 10 (Sales):** IP 192.168.10.x, Subnet Mask 255.255.255.0, **Default Gateway:** 192.168.10.1
- **PC in VLAN 20 (Support):** IP 192.168.20.x, Subnet Mask 255.255.255.0, **Default Gateway:** 192.168.20.1
  Verification: Open the PC command prompt and run ipconfig. Ensure the gateway address matches the exact IP assigned to the router's corresponding sub-interface.

2.Verify Sub-Interface Status and Dot1Q Tags: Step 2: Router Check.

Open the router CLI and check that the sub-interfaces are up and configured with the correct VLAN encapsulation tags:

Plaintext

```
None
show ip interface brief

show running-config
```

If a sub-interface is down or has a mismatched tag (e.g., encapsulation dot1q 10 assigned to a 20.x subnet), reconfigure it:

Plaintext

```
None
configure terminal

interface gigabitethernet 0/0.10

encapsulation dot1q 10

ip address 192.168.10.1 255.255.255.0

exit
```

Verification: Run show ip route to confirm both connected subnets (192.168.10.0/24 and 192.168.20.0/24) appear in the routing table.

3.Verify Switch Trunk Ports and Allowed VLANs: Step 3: Switch Trunk Check.

Ensure the switch port connecting to the router (and all inter-switch trunk links) are active trunks and permit VLANs 10 and 20:

Plaintext

```
None
show interfaces trunk
```

If VLANs 10 and 20 are missing from the allowed list, update the trunk configuration:

Plaintext

```
None
configure terminal

interface fastethernet 0/24

switchport mode trunk

switchport trunk allowed vlan add 10,20

exit
```

Verification: Run show interfaces trunk again and confirm that 10 and 20 appear under "Vlans allowed and active in management domain".

4.Verify VLAN Database Consistency: Step 4: Database Check.

Check that VLANs 10 and 20 exist in the active VLAN database on every switch in the path:

Plaintext

```
None
show vlan brief
```

If either VLAN is missing on a switch, recreate them:

Plaintext

```
None
configure terminal

vlan 10

name Sales

vlan 20

name Support

exit
```

Verification: Confirm both VLAN IDs and their names appear in the show vlan brief output across all switches.

5.Test End-to-End Inter-VLAN Ping: Step 5: Final Verification.

From the PC in VLAN 10 (192.168.10.10), ping the PC in VLAN 20 (192.168.20.20):

Plaintext

```
None
ping 192.168.20.20
```

Verification: Successful replies (Reply from 192.168.20.20: bytes=32 time<1ms TTL=127) confirm that inter-VLAN routing is fully operational.

Set up a basic wireless network using a home Wi-Fi router and connect your laptop or phone to it, then note down the SSID and the default channel used by the router.

ANS:

Setting up a home wireless network and locating your network name (SSID) and default channel involves connecting to your router's web management interface. Here is the step-by-step procedure to set up the router, configure the SSID, check the wireless channel, and connect your device.

1.Connect and Power On the Router: Hardware Connection.

1. Plug the Ethernet cable from your internet modem into the **WAN** (or Internet) port on the back of your Wi-Fi router.
2. Plug the power adapter into the router and switch on the power outlet.
   Verification: Wait 1 to 2 minutes until the power light and WAN/Internet indicator LEDs turn solid green or white.

2.Access the Router Web Interface: Management Access.

1. Connect your laptop or phone to the router's default pre-configured Wi-Fi network (printed on the sticker at the bottom of the router) or plug an Ethernet cable from your laptop into a LAN port.
2. Open a web browser and enter the router's default gateway IP address (commonly 192.168.0.1, 192.168.1.1, or 192.168.8.1).

3. Log in using the administrator username and password found on the router's physical label.
   Verification: The browser successfully loads the router's dashboard or status configuration page.

3.Configure SSID and Check the Default Channel: Wireless Configuration.

1. Navigate to the **Wireless**, **Wi-Fi**, or **WLAN** settings menu in the dashboard.
2. Locate the **SSID** (Service Set Identifier) field and enter your desired wireless network name (for example, Home_Network_2026).
3. Locate the **Channel** or **Control Channel** setting. By default, most routers are set to **Auto**, but if a specific channel is selected, note down the number (such as Channel 6 for the 2.4 GHz band).
4. Save or apply the changes.
   Verification: The router confirms that the wireless settings have been successfully updated.

4.Connect Your Device and Verify: Client Connection.

1. Open the Wi-Fi settings on your phone or laptop and scan for available networks.
2. Select your newly created **SSID**, enter the Wi-Fi security password, and connect.
3. Open a browser or run a command prompt test to confirm internet access.
   Verification: Your device connects successfully and displays an active Wi-Fi connection icon with internet access.

Change your Wi-Fi router's channel setting from 'Auto' to a manually selected channel (such as 1, 6, or 11), then use a Wi-Fi analyzer app on your phone to check if your new channel has less interference from nearby networks.

Hint: Look for overlapping channels and pick one with the least overlap.

ANS:

Switching your Wi-Fi router from "Auto" to a fixed channel—such as **1, 6, or 11** on the 2.4 GHz band—is one of the most effective ways to eliminate co-channel interference from neighbor networks. On the 2.4 GHz spectrum, only channels 1, 6, and 11 are completely non-overlapping; using any other channel (like 2, 3, 4, or 5) causes your traffic to bleed into adjacent frequencies.

Here is how to complete the channel change and verify the results using a Wi-Fi analyzer tool:

1.Change the Wireless Channel: Router Interface.

1. Open a web browser on a connected device and log into your router's administrative dashboard using its gateway IP address (e.g., 192.168.0.1 or 192.168.1.1).

2. Navigate to the **Wireless**, **Wi-Fi**, or **Wireless Settings** menu (often under an "Advanced" or "Basic" sub-tab).
3. Locate the **Channel** dropdown menu (currently set to *Auto* or *Automatic*).
4. Select a non-overlapping channel—**1**, **6**, or **11**—based on your preference, and click **Save** or **Apply**.
   Verification: The router reloads its wireless interface, and your connected devices will briefly disconnect and reconnect to the newly applied channel.

2.Scan with a Wi-Fi Analyzer App: Mobile Verification.

1. On your smartphone, open a Wi-Fi analyzer app (such as *Wi-Fi Analyzer* on Android or *Network Analyzer* on iOS).
2. Switch to the **Channel Graph** or **Channel Rating** view for the 2.4 GHz band.
3. Locate your router's SSID on the graph.
   Verification: Confirm that your network is now broadcasting exclusively on your manually chosen channel (1, 6, or 11) and observe whether neighboring networks are clustered on overlapping frequencies, leaving your selected channel with minimal interference.

Enable WPA2 or WPA3 security on your wireless router, set a strong password, and attempt to connect a device using the new credentials to verify the security setting is active.

ANS:

To secure your wireless network with WPA2 or WPA3 encryption and verify it with a client device, follow these steps:

1.Access Wireless Security Settings: Router Web Interface.

1. Open a web browser on a device connected to your router and enter the router's default gateway IP address (e.g., 192.168.0.1 or 192.168.1.1).
2. Log in using your administrator credentials.
3. Navigate to the **Wireless**, **Wi-Fi**, or **Wireless Security** menu.
   Verification: The security settings configuration page is displayed on your screen.

2.Configure WPA2 or WPA3 and Set a Password: Encryption Selection.

1. Locate the **Security Mode** or **Authentication Type** dropdown menu.
2. Select **WPA2-Personal** (AES) or **WPA3-Personal** (or **WPA2/WPA3 Mixed** if older devices need to connect).
3. In the **Password**, **Pre-Shared Key**, or **Passphrase** field, enter a strong password (at least 12 characters combining uppercase, lowercase, numbers, and symbols).

4. Click **Save** or **Apply**.
   Verification: The router applies the changes and briefly restarts the wireless radio, causing connected devices to disconnect.

3.Reconnect a Device Using New Credentials: Client Verification.

1. On your smartphone or laptop, open the available Wi-Fi networks list.
2. Select your router's SSID. If your device was previously connected, it will prompt you that the saved password is no longer valid.
3. Enter your newly created strong password and click **Connect**.
   Verification: Your device successfully authenticates and establishes an active Wi-Fi connection, confirming that WPA2/WPA3 encryption is active and enforcing security.

Configure MAC filtering on your Wi-Fi router by adding your device's MAC address to the allowed list, then try connecting with another device not on the list and observe the result.

Hint: You can find your device's MAC address in the network settings.

ANS:

To configure MAC address filtering on your wireless router and test access restriction for unauthorized devices, follow these steps:

1.Locate Your Device's Physical Address: Find Device MAC Address.

1. On your smartphone or laptop, navigate to your network settings (e.g., **Settings > About Phone > Status** or **System Settings > Wi-Fi Status**).
2. Locate the **Wi-Fi MAC Address** or **Physical Address** (a 12-character alphanumeric string separated by colons or hyphens, such as A1:B2:C3:D4:E5:F6).
3. Note down this address.
   Verification: The MAC address format is clearly visible and recorded.

2.Access Wireless MAC Filtering Settings: Router Interface.

1. Open a web browser on a device connected to your router and enter the router's default gateway IP address (e.g., 192.168.0.1 or 192.168.1.1).
2. Log in using your administrator credentials.
3. Navigate to the **Wireless**, **Access Control**, or **MAC Filtering** menu (often located under Wireless Advanced settings).
   Verification: The MAC filtering configuration panel is displayed on your screen.

3.Enable Filtering and Add Your MAC Address: Configure Allowed List.

1. Toggle the **MAC Filtering** switch or mode to **Enabled** / **Active**.
2. Set the filtering rule or access mode to **Whitelist** or **Allow Listed** (meaning only devices on the list can connect, blocking all others by default).
3. Add a new entry, paste your device's MAC address, give it a descriptive name (e.g., My_Laptop), and click **Save** or **Add**.
   Verification: Your device's MAC address appears in the active allowed table on the router.

4.Test Connection with an Unlisted Device: Negative Testing.

1. Take a second device (such as another phone or tablet) whose MAC address has **not** been added to the allowed list.
2. Attempt to connect that unlisted device to your Wi-Fi network using the correct password.
   Verification: The unlisted device will fail to obtain an IP address or get stuck on "Connecting / Obtaining IP address," confirming that the router is successfully blocking unauthorized MAC addresses.

Create an Access Control List (ACL) on your router to block access to a specific website (for example, www.youtube.com) from your Wi-Fi network, and test if the block is working by trying to access the site from a connected device.

ANS:

To block access to a specific website like YouTube from your Wi-Fi network using a Cisco router, you can implement a **Named Extended Access Control List (ACL)** combined with Domain Name System (DNS) filtering or IP blocking. Because modern websites frequently change their IP addresses and use Content Delivery Networks (CDNs), the most reliable method on a standard router is to block traffic destined for the domain's resolved IP addresses or restrict traffic using Layer 4 port filtering.

Here is the step-by-step procedure to configure the ACL, apply it to the internal Wi-Fi interface, and verify the block:

1.Resolve the Target Website IP Addresses:DNS Resolution.

Before creating the ACL, identify the public IP addresses associated with the target website (www.youtube.com). From a computer connected to your router, run a DNS lookup:

Plaintext

```
None
nslookup www.youtube.com
```

Verification: Note down the returned IP addresses (e.g., 142.250.190.46 or the broader subnet block).

2.Create the Extended Named ACL: Global Configuration.

Open the router CLI, enter global configuration mode, and define an extended ACL that denies outbound TCP/UDP traffic to the target website's IP addresses while permitting all other traffic:

Plaintext

```
None
enable

configure terminal

ip access-list extended BLOCK_YOUTUBE

deny tcp any host 142.250.190.46 eq www

deny tcp any host 142.250.190.46 eq 443

permit ip any any

exit
```

Verification: Run show access-lists BLOCK_YOUTUBE to confirm the entries are correctly recorded in the router's memory.

3.Apply the ACL to the LAN/Wi-Fi Interface: Interface Application.

Apply the ACL to the interface facing your internal Wi-Fi users (incoming direction) so it filters traffic originating from your wireless network:

Plaintext

```
None
interface gigabitethernet 0/1.10

ip access-group BLOCK_YOUTUBE in
```

```
exit
```

Verification: Run show ip interface gigabitethernet 0/1.10 and verify that the output displays the active inbound access list BLOCK_YOUTUBE.

4.Test the Block from a Connected Device: Client Verification.

1. Connect a smartphone or laptop to your Wi-Fi network.
2. Open a web browser and attempt to navigate to https://www.youtube.com.
   Verification: The browser times out or displays a connection error (such as "Site can't be reached" or connection refused), confirming that the ACL is successfully blocking traffic to the target site while allowing normal internet browsing on other websites.